SAFE Quantum Machine Learning with Variational Quantum Classifiers

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Motivation & Contribution

- High-stakes ML (e.g., medical imaging) requires models which imply:
  - Robustness to noise and perturbations (security/stability)
  - Stability under feature removal (explainability robustness)
  - Consistent behavior across operating conditions
- Quantum models provide structured inductive biases:
  - Normalized representations (amplitude encoding)
  - Bounded and smooth transformations
- **Goal:** compare quantum and classical models under a unified SAFE framework

Literature Review

Approach	Examples	Strengths	Limitations
Classical Deep Learning	CNNs, ResNet, nnU-Net	High accuracy in medical imaging	Brittle to noise, weak robustness and explainability
Classical ML on Deep Features	SVM, RF, MLP	Strong performance with fewer parameters	Reliability not jointly evaluated
Hybrid Quantum–Classical ML	VQC-based models	Compact, constrained representations	SAFE properties rarely analyzed
SAFE/Reliability Metrics	RG, SAFE-AI metrics	Unified accuracy, robustness and explainability	Not applied to QML

Quantum Machine Learning Model (I)

- Hybrid **deep** + **quantum** pipeline for SAFE-oriented learning
- **Deep feature extraction (frozen ResNet-18)**
  - MRI image $I \rightarrow$ 512-D embedding

$$\mathbf{z} = f_\theta(I), \quad \mathbf{z} \in \mathbb{R}^{512}$$

  - Replace final FC layer with identity; extract features in eval mode
  - Keep full 512-D representation (no PCA / dimensionality reduction)
- End-to-end training for classification (cross-entropy) with cross-validated evaluation

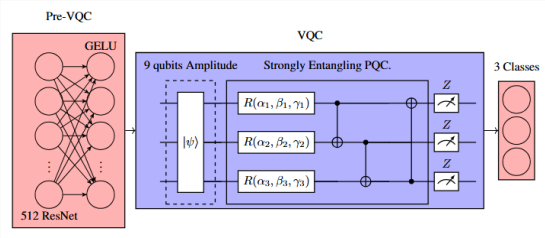
Quantum Machine Learning Model (II)

- **Pre-VQC classical layer**
  - Linear + GELU adaptation
  - Stabilizes amplitude encoding (ablation degrades training)
- **Amplitude encoding (9 qubits)**

$$\mathbf{x} = \frac{\mathbf{z}}{\|\mathbf{z}\|}, \qquad |\psi(\mathbf{x})\rangle = \sum_{i=0}^{511} x_i |i\rangle$$

- **VQC + measurement**
  - Strongly entangling layer (rotations + CNOTs)
  - $q_i = \langle Z_i \rangle \in [-1, 1]$
- **Classifier head**

$$y = W_c q + b_c$$



SAFE-AI Metrics

- All SAFE-AI metrics derive from the Cramér–von Mises divergence between empirical CDFs
- **Rank Graduation (RG):**

$$\mathrm{RG}(Y, Y') = 1 - \frac{\mathrm{CvM}_1(F_Y, F_{Y'})}{G(Y)}$$

- **Special cases:**
  - Binary: $\mathrm{RG} = \mathrm{AUC}$
  - Multiclass: one-vs-rest, weighted by class frequency
  - Continuous: complementary to predictive R^2
- **SAFE dimensions: RGA** (accuracy), **RGR** (robustness), **RGE** (explainability)

Experimental Setup

Dataset

- Brain tumor MRI classification
- 3 classes, 6,056 images

Models

- Linear (Logistic Regression)
- Random Forest
- SVM
- MLP
- Variational Quantum Classifier

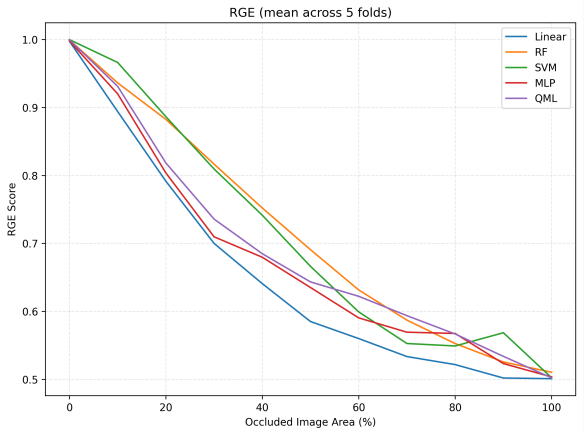
Feature representation

- ResNet-18 embeddings (512-D)
- Same features for all models

Evaluation

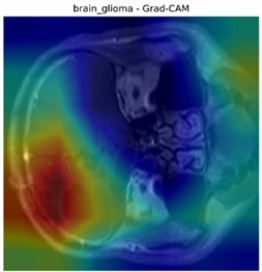
- 5-fold cross-validation
- Accuracy, F1, MSE
- SAFE: RGA, AURGA, AURGR, AURGE

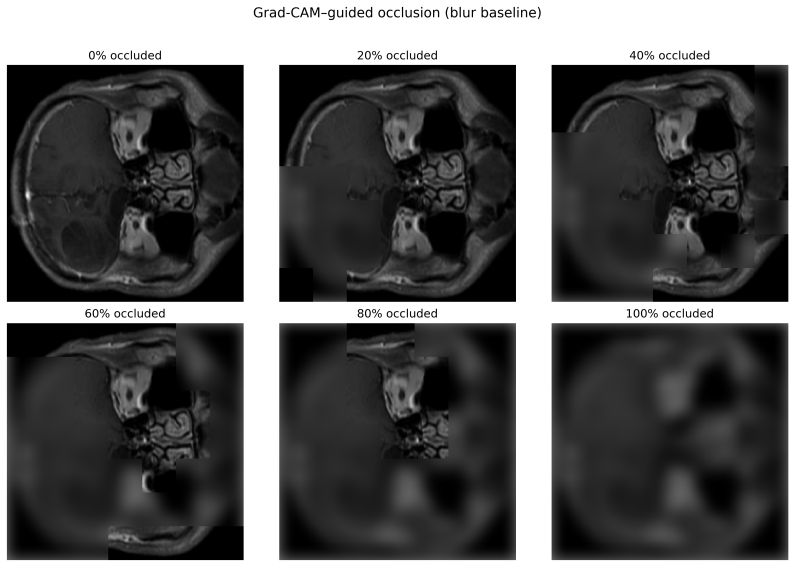
SAFE Results (I)



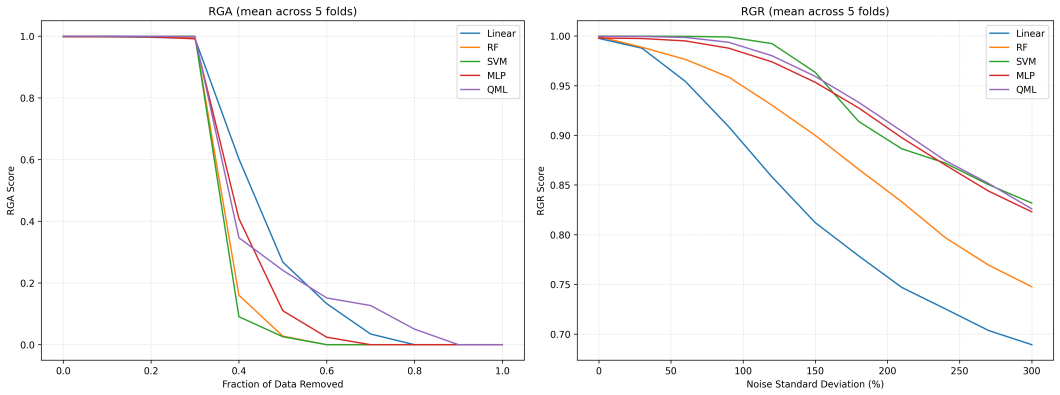
- **Input-level explainability**
  - MRI images directly occluded
- **Relevance ordering**
  - Grad-CAM on frozen ResNet-18 probe
  - Patch relevance = mean Grad-CAM score
- **Progressive occlusion**
  - Non-overlapping $p \times p$ patches
  - Blur top-K relevant patches
- **RGE computation**
  - Compare predictions before/after occlusion
  - Stability under informative region removal

RGE Occlusion Process





SAFE Results (II)

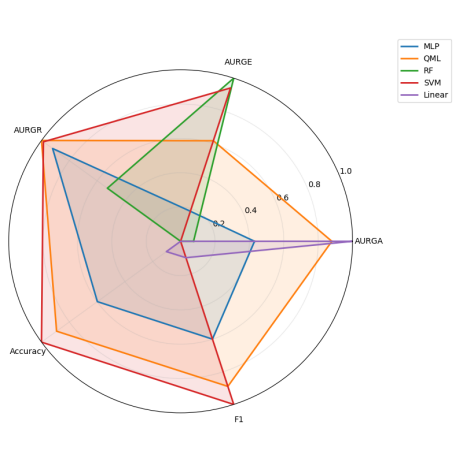


Model Comparison (5-fold CV)

Model	Acc	F1	MSE	RGA	AURGA	AURGR	AURGE
Linear	0.938	0.938	0.034	0.9976	**0.452**	0.832	0.648
RF	0.933	0.933	0.067	0.9994	0.368	0.889	**0.712**
SVM	**0.983**	**0.983**	**0.008**	**0.9999**	0.361	0.939	0.709
MLP	0.963	0.963	0.020	0.9988	0.400	0.932	0.660
QML	0.977	0.977	0.013	0.9996	0.441	**0.940**	0.688

- **SVM dominates traditional metrics** (Accuracy, F1, MSE), but SAFE reveals hidden trade-offs
- **QML**: strong performance with best noise robustness (AURGR $= 0.940$)
- Each model excels in different SAFE dimensions $\rightarrow$ balanced reliability matters

Balanced SAFE Profile (Radar plot)



- **VQC**: most balanced SAFE profile
- RF: strongest AURGE, weak elsewhere
- SVM: best Accuracy/F1, low AURGA (resilience to data removal)
- Linear: strong AURGA

Conclusions

- Proposed hybrid classical–quantum model: deep features + amplitude encoding + VQC
- Competitive predictive performance on brain tumor MRI classification
- SAFE-AI metrics reveal advantages beyond accuracy (robustness and explainability stability)
- Quantum models should be viewed as **stability-oriented learners**, not only accuracy boosters
- SAFE-AI enables unified evaluation of classical and quantum models in safety-critical settings